

### LESSON 311 (1.5 DUAL 1.3 INSTRUMENT)

#### Lesson Objective

During this lesson, the student continues to develop the knowledge and skill with respect to attitude instrument flying using VOR, DME and NDB navigation equipment. Engine failures will be introduced.

#### Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
  - FIT procedures for VOR Intercepting and Tracking.
  - FIT procedures for Intercepting and Tracking DME Arcs from radials and radar vectors.
- Student questions.

#### Lesson Content (Ground)

##### Go/no-go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly-away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

#### Lesson Content (Flight)

##### Aircraft Checklist Items

*Emphasize timeliness to minimize ground time!*

- Instrument Cockpit Check
  - Student Tells, Student Does.
    - Have the student read each item from the checklist out loud.
    - Discipline the student to set bugs, bearing pointers, and any auxiliary features.
  - The autopilot check can be conducted now:
    - Engage autopilot.
    - Test roll: left and right.
    - Test pitch: up and down (VS mode).
    - Ensure autopilot can be overridden.

- Check the red disconnect button.
- Electric trim test:
  - Check the motion of the trim wheel.
  - Check the S-TEC trim annunciators.
  - Each half-switch should not move the trim wheel.
  - The disconnect switch should stop the wheel.
- Before taxi check:
  - Student Tells, Student Does.
    - Airspeed indicator (PFD and standby): zero.
    - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
    - Standby attitude indicator: standby battery check and adjust the miniature aircraft.
    - Altimeter (PFD and standby): +/- 75 feet of station elevation.
    - Vertical speed indicator (PFD): reads 0 or use current indication.
    - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
    - HSI: matches compass and turns appropriately.
    - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

### Departure

- Follow the tower's instructions for departure.
- The student must put the hood or foggles on when safe.
- Climb out:
  - Student Does, Instructor Evaluates.
    - Give vectors to the practice area.
    - Maintain  $V_Y$ .
    - Conduct the 1,000ft check.
    - Request nonstandard constant rate and airspeed climbs.
    - Level off at a given altitude.

### VOR Intercepting and Tracking

- Ensure that the student can TUNE and ID the VOR.
- Intercepting and Tracking Within 5NM:
  - Student Does, Instructor Evaluates.
    - Specify a radial for the student to intercept outbound.
    - Check if the student notes DME.
    - Ensure a proper intercept angle of 30 - 45 degrees.
    - Course must be corrected for wind.
- Airway Intercept and Tracking:
  - Student Does, Instructor Evaluates.
    - Beyond 5nm, from the current radial, intercept airway.

## Instrument Rating Course

- Ensure the trainee identifies the correct radial.
- Evaluate procedure and intercept heading.
- Intercepting and Tracking Outside 5NM:
  - Student Does, Instructor Evaluates.
    - Give the student another radial to intercept outbound from the VOR.
    - Ensure that the student notes the difference and doubles.
    - Ensure that the student is twisting the desired course before applying any intercept.
    - Guard the student against using  $<20^\circ$  or  $>90^\circ$ .
    - Have the student bug the intercept heading and intercept the radial.
    - Repeat the above steps as time permits, challenging the student with Inbound to Outbound or Outbound to Inbound turns.

### Intercepting and Tracking DME Arcs

- Intercepting and Tracking from a Radial:
  - Student Does, Instructor Evaluates.
    - Note the radial, direction, and distance from the station.
    - Give ARC Instructions with Compass Direction, Distance, and the Fix.
    - Make sure the student understands the instruction.
    - The no-wind intercept is  $100^\circ$  to the present radial (Advance OBS  $10^\circ$  and add  $90^\circ$  in direction of the arc).
    - Join the arc at 0.5 nm prior to the arc (depending on headwind/tailwind) – immediately advance OBS  $10^\circ$
    - Make sure OBS is set up in the FROM indication.
    - If undershooting the arc, roll out early; If overshooting, continue to turn to intercept the arc.
    - Let the student choose what method they want to use to track DME arcs.
- Intercepting and Tracking a DME Arc from Radar Vectors:
  - Student Does, Instructor Evaluates.
    - Give the student a radar vector and an ARC distance.
    - Ensure that the student, at 1.5 DME prior to the arc, notes the radial, and calculates the intercept heading.
    - The no-wind intercept is  $100^\circ$  to the present radial (or  $90^\circ$  to the succeeding radial).
    - Join the arc at 0.5 nm prior (depending on degrees to be turned and winds) to the arc or as necessary to prevent overshoot/undershoot.

### Unusual Attitudes

- Recognition and recovery from unusual attitudes:
  - Student Does, Instructor Evaluates.
    - Conduct the pre-maneuver checklist.
    - Disorient the student - use slips and skids.

- Prompt recovery.
- On the last unusual attitude, dim the PFD.
- Prompt recovery (ensure that the student is now using the standby instruments).
- Resume partial panel straight and level flight.

### GPS Intercepting and Tracking

- Direct-To-Navigation:
  - Instructor Tells, Student Does.
    - Select a reasonably nearby waypoint to navigate to.
    - Coach the student on the Direct key or the NRST menu.
    - Stress that the CDI needs to be set up to track GPS, NOT VLOC, with the CDI Key.
    - Ensure that the PFD is set to the GPS course.
    - Now, the autopilot GPSS mode can be selected by pressing NAV twice.
    - Show the student how to determine GPS mode and sensitivity.
- OBS Mode:
  - Instructor Tells, Student Does.
    - Coach the student on the use of OBS Key.
    - This key will allow 360 different courses to be set – effectively “turning a GPS waypoint into a VOR”.
    - GPS courses can even be intercepted and tracked using the exact same procedures as with VORs.
    - DME Arcs can even be based on GPS waypoints using the CDI, HSI, or RMI.

### Engine Failure During Cruise

- Simulated Engine Failure in IMC:
  - Student Tells, Student Does:
    - Instructor reduces power to idle.
    - The student will conduct checklists.
    - The student will use NRST and request emergency vectors.
    - (Optional) Allow the student to glide to a Non-Towered Airport and land.
    - Recover from the emergency as appropriate.

### Localizer Intercepting and Tracking

- Vectors Inbound:
  - Student Does, Instructor Evaluates.
    - Ensure the student can correctly set up for the approach:
      - Weather (obtain the ATIS).
      - Instruments (load the procedure, tune and ID the localizer, set up the PFD to the correct NAV source, and set the approach course).
      - Radios (set COM1/COM2).
      - Enroute brief (this step can be skipped in stage 1).
      - Descent and arrival checklist.
    - Follow ATC instructions.

- Flaps should be added to 10 degrees prior to joining the localizer.
- Stress to the student to turn early once the needle leaves full-scale deflection.
- Evaluate the size of corrections and deflections.
- Cue the student on descending to MDA (or track the glideslope if they can).
- Prior to the runway threshold, allow the student to go visual.
- Land and return to FIT.

### Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 312a.
- Assign Homework:
  - Study for the mock!

### Completion Standards

At the completion of this lesson, the student should:

1. Be able to demonstrate increased proficiency in VOR, DME, localizer, and NDB navigation with minimal instructor assistance.